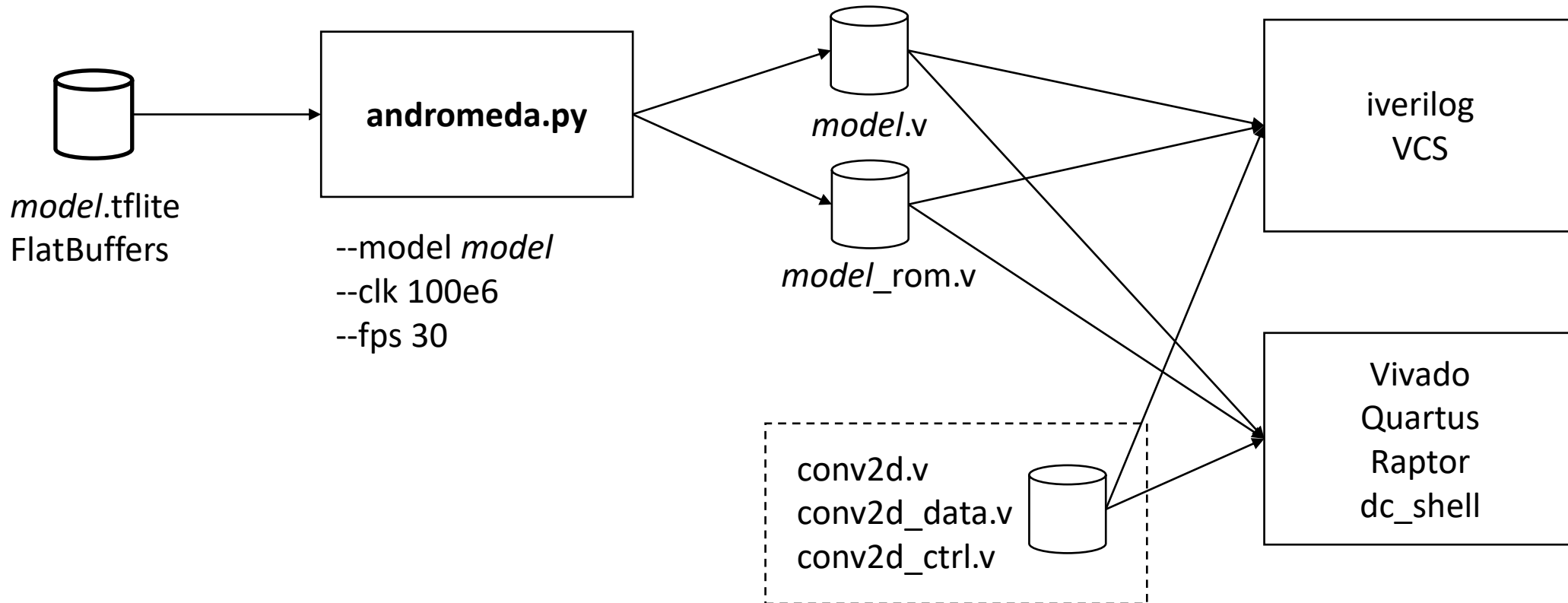
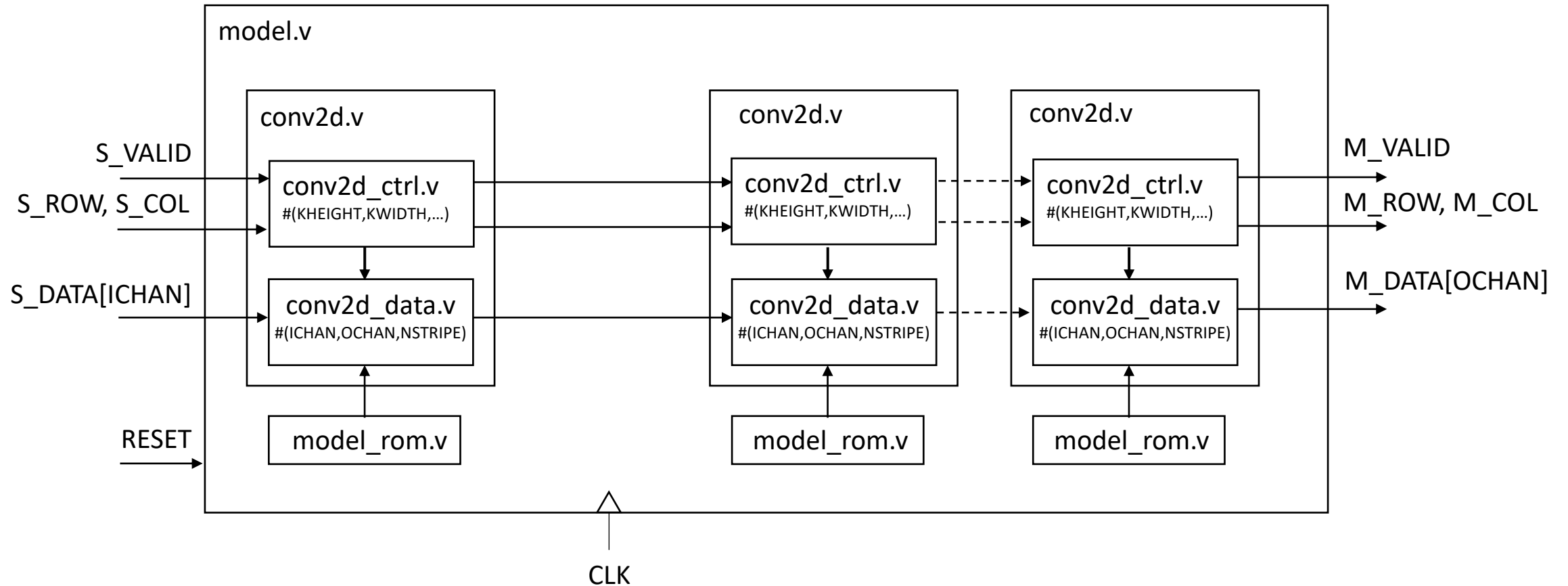


Streaming CNN compiler

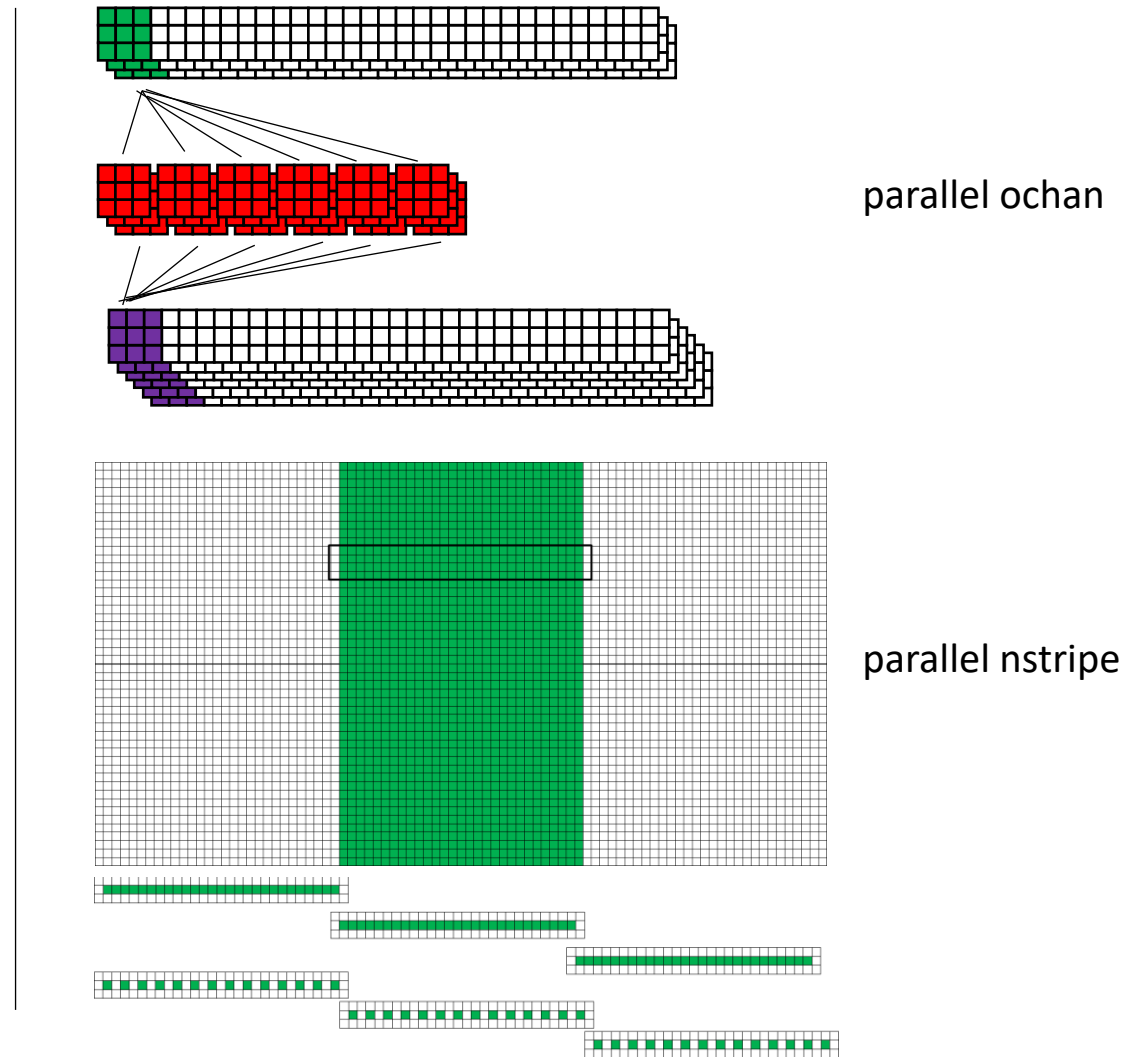
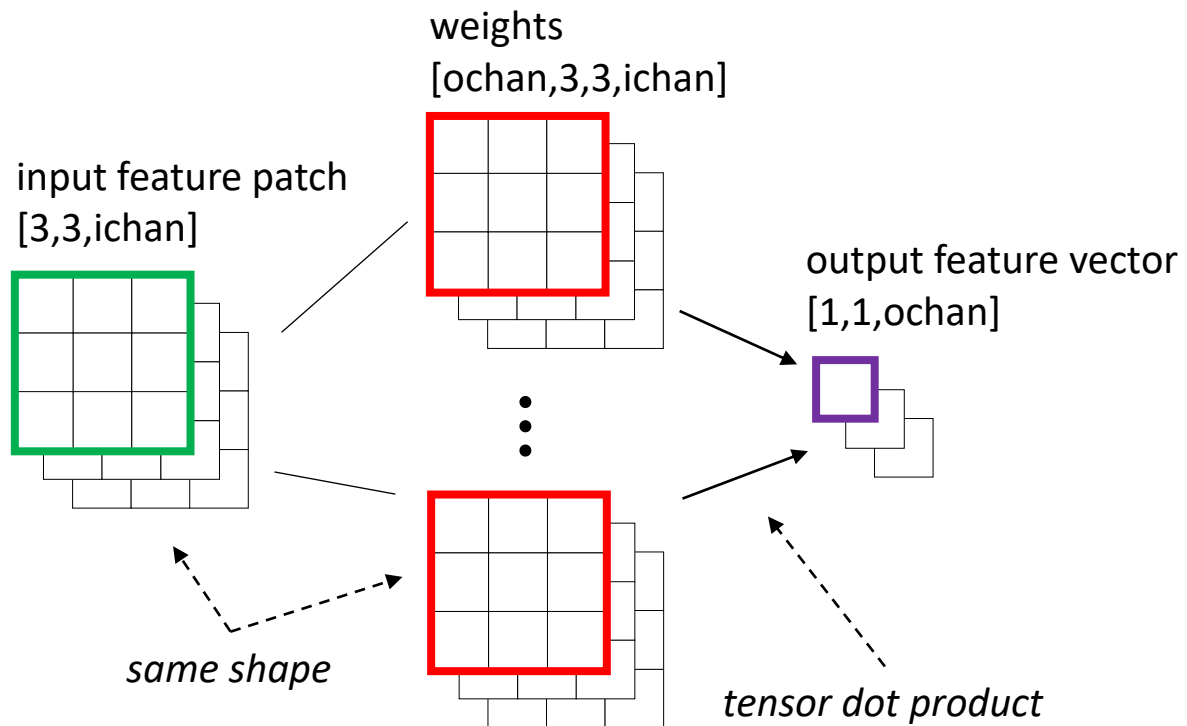


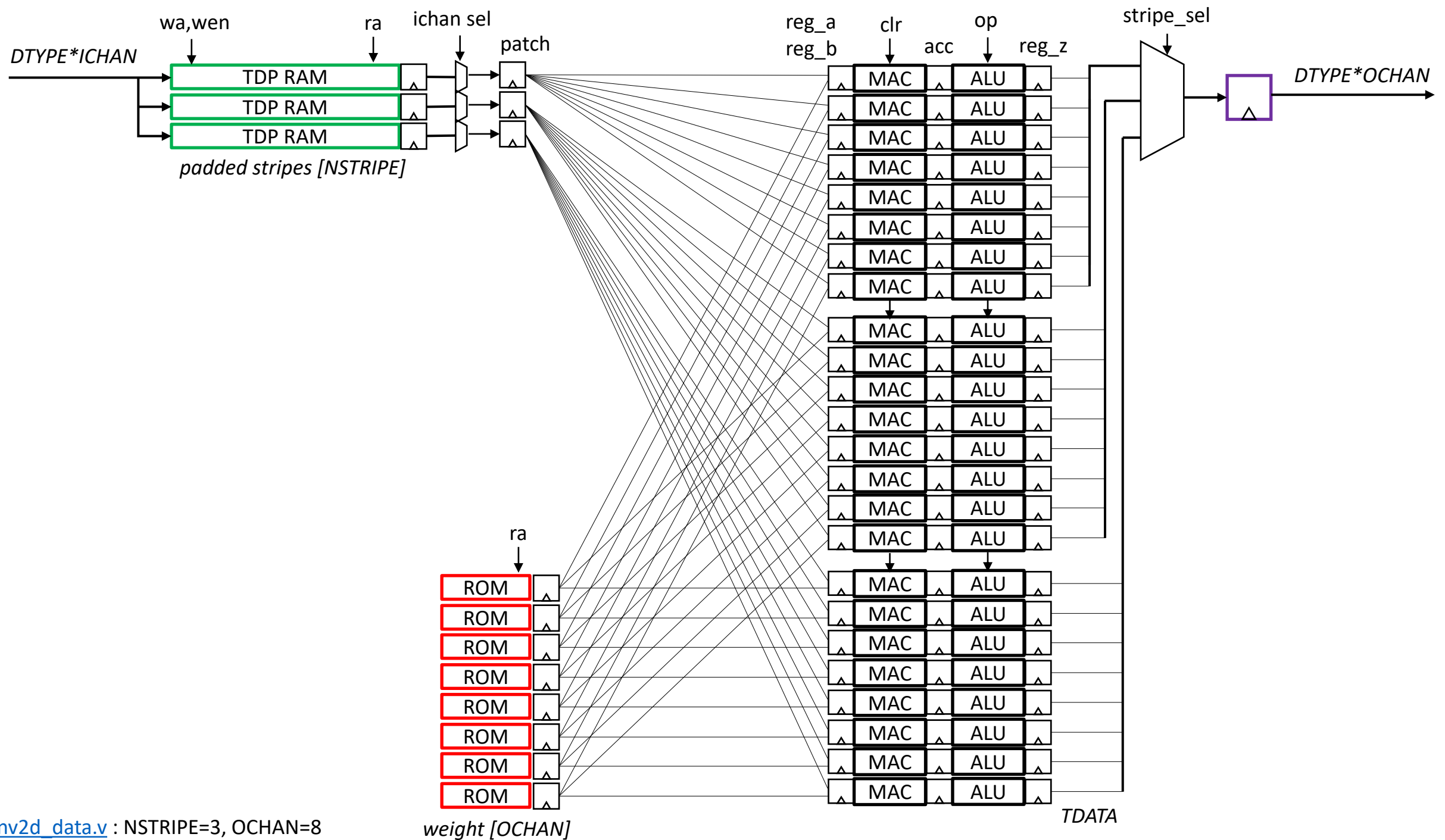
automated tool to compile TFLite model into synthesizable Verilog, using conv2d.v parameterized module
compiler generates parallel code for $OCHAN * NSTRIPE$ MAC operations per layer

Generated Verilog

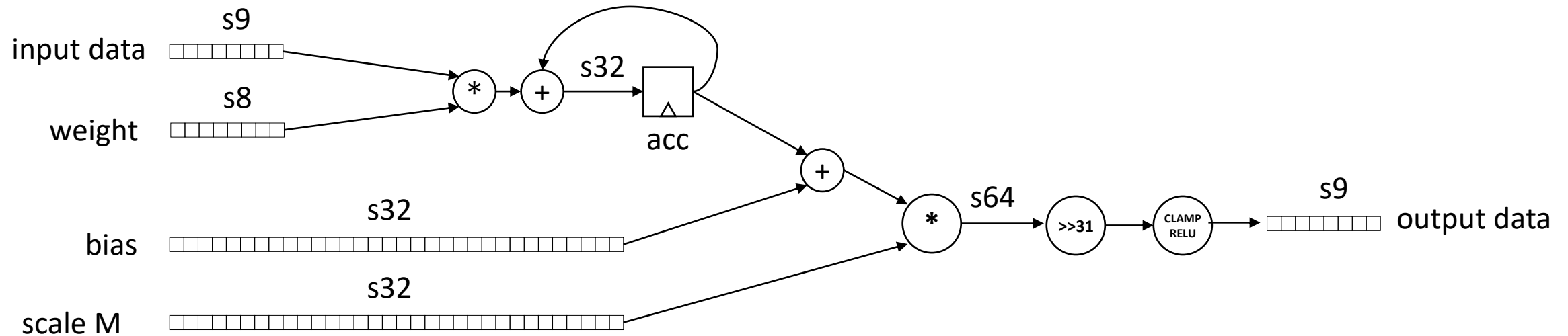


Parallel computation of CNN dot products





TFLite int8 quantization



- 1: compute dot product ($s9 * s9 + s32$)
- 2: add bias ($s32 + s32$)
3. multiply by scale ($s32 * s32$)
4. shift right $\gg 31$
5. clamp to s9 bits and RELU

CNN striped parallel execution

